

Practice Data Prompt

Practice Round Prompt

We are interested in expanding the use of our GPS data. Using the dataset provided, analyze race outcomes using both traditional (non-GPS) data and GPS data variables. Assess how the inclusion of GPS data changes, refines, or enhances the interpretation of race performance relative to traditional race metrics. Specifically, focus on whether GPS metrics (stride, running_time, distance_ran..etc) can better predict finishing position/ horse performance than traditional point-of-call data alone. Highlight any new performance insights that can be derived from the GPS data and evaluate their analytical significance.

Tabs Structure

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The '**GPS Races for LRL Practice Round**' has been created to get you familiar with the data that exist in our database, highlighting the races that do have GPS data. In the spreadsheet, three tabs exist,

1. **Laurel Park GPS races (past 60days)**

- This tab presents the past 60 days of racing activity at Laurel Park Track . A total of 148 races ran at Laurel Park (Track code is LRL) in the past 60 days. In each race, each horse has multiple rows present per race as the horse's movement gets collected at different gates within the race. Gate 0 is the final gate in the race. The number of gates that exist per race depend on the race distance.

2. **Starters pps (relating to Laurel Park GPS Races)**

- For all the horses that took part in any of the 148 races at LRL in the past 60 days, their 3 most recent past performances/races (PPs) were queried out and put in this tab. Horses that do not have 3 PPs (due to them not racing much yet), will show the 1 or 2 PPs that exist for them. It is important to note too that for some horses, the LRL race they took part in in the past 60 days was their first start ever, causing no PPs to exist for them at all in this tab. There are also cases where some horses might have taken part in more than one of the 148 LRL park races, more than 3 PPs for that horse might exist in this tab as the 3 most recent PPs for the horse depends on which race came before which.
- In this tab, you will also find the non-GPS data we collect for the races. For background, as it stands right now, not all the tracks are equipped to collect GPS data, causing some races to have both GPS & non-GPS data and some other races to only have non-GPS data, which is a limitation we have in this regard. To be able to quickly get an idea of which PPs have GPS data and which don't, please refer to the gps_data column. If GPS data exists for the listed PP, an X is denoted, otherwise it is left blank.

3. **GPS PPS**

- For any of the PPS (listed in the Starters PPs) that did have GPS data, that GPS data for the race can be found in this tab here. As a reminder, similar to the first tab, each horse has multiple rows present within each race due to the different gates that exist.

Variables Definitions

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The following goes over some codes and definitions. For the most part, a lot of the variable definitions can be found here: [Codes and Definitions](#). Any variables not explained in that page and/or GPS-related variables, can be found below:

1. **Distance_id:** the race distance as reported in the database
2. **Distance_unit:** indicating the unit of the race distance_id reported. F indicates Furlongs.
3. **About_distance_indicator:** this column will either have a value of A or blank. For example a race of 6.5 F with the 'A' value in the about distance indicator indicate that the race was **about** 6.5 furlongs (not exactly 6.5 furlongs)
4. **Distance:** When the race chart gets published, this column is what gets published. The main difference between it and the distance_id is the unit the distance is reported in. For example, a race distance of 8.5 furlongs (reported in database) equals to 1 1/16 Miles (the published value)
5. **Surface:** D indicates horses ran on dirt. T indicate horses ran on Turf
6. **Grade:** this column only relates to STK race types. Only 1,2, or 3 can be reported here. With 1 being the highest Graded Stakes race, 2 being second highest and third indicating the third highest graded stakes. Generally speaking, Grade 1 Stakes races often represent the highest level of Thoroughbred horse racing.
7. **Purse:** the prize money set for the race. It is usually split among the horses depending on where they finish in the race.
8. **Post Position:** The position the horse is placed at in the starting gate. With 1 indicating the horse is starting closest to the rail.
9. **Gate:** represent the different distance markers throughout the race used to capture the gps data. The highest gate # in a race represent the earliest gps data collection point in the race. Gate 0 indicates the final gps data collection point. In each GPS race, the horse will have rows for each of the gates available for the race. The number of gates in the race depends on race distance.
 - The gates are set to be 1/16th of a Mile away from each other for the races. 1/16th of a Mile = .5 furlong. That in mind, for example, an 8 furlong race will have the highest gate distance marker be 7.5 furlong, which is only .5 furlongs (1/16th mile) from the beginning of the race. This is the case for the majority of the races.

For cases where the race distance is 8.32 furlongs (which is equal to the published value of 1M 40Y), since the .32 furlongs is not a full .5 furlongs, the gates are split differently with the highest/initial gate being set at 7.68 furlongs and so on.

- 10. Position:** this represents the placement of the horse during that gate check i point.
- 11. Official_position:** represent where the horse placed at the end of the race.
- 12. Sectional_time:** represents the time it took the horse to finish every section (1/16th of a mile in most cases, except in the few cases mentioned above of the yard race cases) present in the race.
- 13. Running_time:** same as sectional_time column. The only difference is that in this column, it is more of up until this point- how long has the horse been running for. For example, say a race has 6.5 gates in the race and at gate 6.5, the horse's sectional time is 7.45 and at gate 6, his sectional time was 5.51. In the running time for this horse, at gate 6.5, his running time is 7.45. At gate 6, his running time is 13 (7.49+5.51) and so on.
- 14. Time_behind:** how much slower was the horse from the leader at that gate (in seconds)
- 15. Distance_behind:** the distance (reported in meters) of how behind was the horse from the leader at that gate.
- 16. Distance_ran:** the distance (reported in meters) the horse ran in that section. In other words, the distance the horse ran to finish each 1/16th of a Mile. This differs from horse to horse as some horses might run more closely towards the rail making them run less distance, while other horses are running further away from the rail, causing them to run more distance.
- 17. Cumulative_distance_ran:** same as how sectional time and running_Time are set up, except this time focus is on distance_ran and cumulative_distance_ran is just adding on the distance completed as the race progresses.
- 18. Strides:** represent the number of strides the horse took during that section/gate
- 19. Cumulative_strides:** same case as distance and cumulative distance columns but this relates to the strides portion.
- 20. Position_at_point_of_call_1 | position_at_point_of_call_2 | position_at_point_of_call_3 | position_at_point_of_call_4 | position_at_point_of_call_5:**

These columns are traditional non-GPS data points indicating the position of the horse at the different point of calls. The different point of calls that exist for each race distance can be found here: [Point of calls](#)

21. Length_behind_at_poc_1 | length_behind_at_poc_2 | length_behind_at_poc_3 | length_behind_at_poc_4 | length_behind_at_poc_5:

These columns are traditional non-GPS data points indicating how far behind the horse is from the horse in front of it at the different point of calls listed in the link above. A horse at the first place at the different pt of calls will have a 0 in that column as no horses are in front of him. A value of 9999 indicate that the horse did not finish the race
The values indicated in this column are measured in horse lengths:

It's a cumulative value where
100 is 1 horse length.
5 is a nose,
10 a head,
25 a neck/quarter length,
50 a half a horse,
75 is 3/4 of a horse length

22. Length_ahead_at_poc_1 | length_ahead_at_poc_2 | length_ahead_at_poc_3 | length_ahead_at_poc_4 | length_ahead_at_poc_5:

These columns are traditional non-GPS data points indicating how much further ahead the horse is from the one behind it. It uses the different point of calls listed in the link above. A horse at the last place at the different point of calls will have a 0 in that column as no horses are behind him. A value of 9999 indicate that the horse did not finish the race. The values indicated in this column are measured in horse lengths as well (the ones listed above)

23. Field_size: number of horses running in the race